

KUNAL RAI

Full-Stack Developer | Real-Time Systems | AI Applications

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Professional Summary

Final-year Computer Science undergraduate (GPA 8.13/10) with 1.5+ years of hands-on project experience in Full-Stack Development (React, Next.js, Node.js, FastAPI) and Agentic Generative AI (LangChain, LangGraph, RAG, Qdrant, LangSmith). Experienced in System Design through microservices, event-driven architecture (RabbitMQ), Redis caching, and CI/CD with Docker. Strong DSA and problem-solving background, ranked top 11% (4,509/40,000) in a global coding contest. Seeking SDE / Generative AI Engineer roles.

Education

Bennett University

Bachelor of Technology in Computer Science **GPA: 8.13/10**

Sep 2023 – Sep 2027

Greater Noida, Uttar Pradesh

Relevant Coursework

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|--------------------------|---------------------------|-----------------|------------------------|
| • Data Structures | • Database Management | • FullStack | • Computer Networks |
| • Analysis of Algorithms | • Artificial Intelligence | • System Design | • Software Engineering |

Projects and Applied Experience

EKLAVYA – AI Learning Platform | *Next.js, LangGraph, FastAPI, LangSmith, MongoDB, Redis, RabbitMQ, Qdrant, Docker*

- Architected **5 autonomous LangGraph agents** (CRAG, Quiz Remediation, Curriculum, Study Session, Essay Grader) with conditional branching, **Pydantic-validated** self-correction loops, and parallel fan-out topologies.
- Built a **Corrective RAG (CRAG)** pipeline: **Qdrant HNSW** + courseId payload-filtered retrieval, **Tavily** fallback on low-relevance chunks, and a hallucination-grader node for verified factual grounding.
- Persisted multi-turn tutoring state via **LangGraph MemorySaver** checkpoints (thread_id-keyed), tracking topic history and comprehension server-side.
- Parallelized essay grading across **3 isolated rubric nodes** (Accuracy, Completeness, Clarity) to eliminate prompt dilution and boost scoring consistency.
- Built a **3-tier LLM router** with quota-fallback, plus **LangSmith** observability: metadata tagging, per-run feedback logging.
- **GitHub:** github.com/VampKunal/EKLAVYA-V2 **Live:** eklavya-v2.vercel.app

IdeaRoom – Real-Time Collaborative Whiteboard | *Next.js, Node.js, Socket.IO, MongoDB, Redis, RabbitMQ, Docker*

- Built a **real-time collaborative whiteboard** using **Socket.IO** and integrated the **Gemini API** to convert natural language prompts into structured flowcharts for collaborative editing.
- Architected a **4-service microservices backend** (API Gateway, Auth, Collaboration Engine, Snapshot Worker), enabling independent scaling and improving service fault isolation.
- Improved data retrieval performance by **40%** by introducing **Redis caching** over MongoDB while offloading snapshot writes through **RabbitMQ event queues**.
- Containerized backend services using **Docker** and deployed production infrastructure.
- **GitHub:** github.com/VampKunal/IdeaRoom **Live:** idea-room-ashy.vercel.app

Technical Skills

Languages: C++, Python, JavaScript, TypeScript

Frontend: HTML5, CSS3, React, Next.js, Tailwind CSS, Zustand, React Query

Backend: Node.js, Express.js, REST API Design, NestJS, FastAPI, WebSockets (Socket.IO)

Databases: MongoDB, PostgreSQL, Redis

Generative AI: LangChain, LangGraph, RAG, Prompt Engineering, Vector Embeddings, Vector Databases (Qdrant, pgvector), OpenAI API, Google Gemini API, AI Agents

Tools: Docker, RabbitMQ, Git, GitHub Actions (CI/CD), Railway, Vercel

Problem Solving & Competitive Programming

- Ranked in the **Top 11% (4509 / 40,000 participants)** in a global competitive programming contest.